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Machine Learning for 1-Year Graft Failure Prediction in Lung Transplant Recipients: The Korean Organ Transplantation Registry

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ABSTRACT

BACKGROUND: In regions with limited donor availability, optimizing efficiency in lung transplant decision-making is crucial. Preoperative prediction of 1-year graft failure can enhance candidate selection and clinical decision-making.

METHODS: We utilized data from the Korean Organ Transplantation Registry to develop and validate a deep learning-based model for predicting 1-year graft failure after lung transplantation. A total of 240 cases were analyzed using 5-fold cross-validation. Among 25 preoperative factors associated with 1-year graft failure, we selected the top 9 variables with coefficients ≥ 0.25 for model development.

RESULTS: Of the 240 lung transplant recipients, 55 (22.92%) developed graft failure within 1 year, while 185 survived. The final predictive model incorporated nine key pretransplant factors: age, bronchiolitis obliterans syndrome after hematopoietic cell transplantation, pretransplant bacteremia, bronchiectasis, creatinine, diabetes, positive human leukocyte antigen crossmatch, panel reactive antibody 1 peak mean fluorescence intensity, and pretransplant steroid use. The multilayer perceptron model demonstrated strong predictive performance, achieving an area under the curve of 0.780 and an accuracy of 0.733.

CONCLUSIONS: Our machine learning-based model effectively predicts 1-year graft failure in lung transplant recipients using a minimal set of pretransplant variables. Further validation is needed to confirm its clinical applicability.

Abbreviations: BRF, balanced random forest; GBM, gradient boosting machine; KOTRY, Korean Organ Transplantation Registry; LR, logistic regression; LTx, lung transplantation; ML, machine learning; MLP, multilayer perceptron; RF, random forest; SVM, support vector machine.

Dasom Noh and Sunyoung Kwon contributed equally to this study.

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1 | Introduction

Lung transplantation (LTx) is a life-saving treatment for patients with end-stage lung disease. Despite advances in surgical techniques and postoperative care, early mortality remains a significant challenge, with 1-year survival rates varying considerably among different cohorts. Globally, 1-year survival rates exceed 80%, with high-volume centers reporting rates as high as 90% [1–3]. However, in South Korea, the 1-year survival rate for lung transplantation has only recently begun to exceed 70% [4].

Accurately predicting 1-year mortality following LTx is crucial for improving patient outcomes, optimizing organ allocation, and guiding clinical decision-making. Traditional statistical models have been widely used to identify risk factors and predict outcomes in LTx recipients. However, most of these models are based on data from the United States or Europe, which limits their applicability to South Korean patients [5]. Furthermore, these models often rely on linear assumptions and may not fully capture the complex, nonlinear interactions between clinical variables.

In recent years, machine learning (ML) has emerged as a powerful tool in medical research, offering the ability to analyze large datasets and uncover intricate patterns that might be overlooked by conventional methods [6]. ML models can handle high-dimensional data, account for nonlinear relationships, and improve predictive accuracy, making them particularly suitable for survival analysis in organ transplantation.

In South Korea, the Korean Organ Transplantation Registry (KOTRY) provides a comprehensive, multicenter dataset of LTx recipients, offering a unique opportunity to develop and validate predictive models specifically tailored to the South Korean population. We leveraged this dataset to develop and validate a ML-based model to predict 1-year graft failure in LTx recipients.

2 | Methods

The KOTRY is a prospective multicenter cohort registry encompassing lung, kidney, liver, pancreas, and heart transplantations performed in South Korea. This study utilized data from the KOTRY LTx cohort registered between March 2015 and December 2021. During the study period, a total of 303 patients were initially enrolled. Patients who withdrew from the registry, underwent heart-lung transplantation, were lost to follow-up, or had incomplete data were excluded from the study. After applying these exclusion criteria, 240 patients remained eligible and were included in the final analysis (Figure 1). In cases where retransplantation was performed due to graft failure, the date of retransplantation was considered equivalent to the date of death. This study was approved by the Institutional Review Board of Pusan National University Yangsan Hospital (IRB No. 04-2015-012, 05-2021-088). Written informed consent was obtained from all participants prior to LTx for enrollment in the KOTRY registry.

2.1 | Clinical Variables and Definitions

Pretransplant bacteremia was defined as any bloodstream infection confirmed by clinical evidence of active infection and a

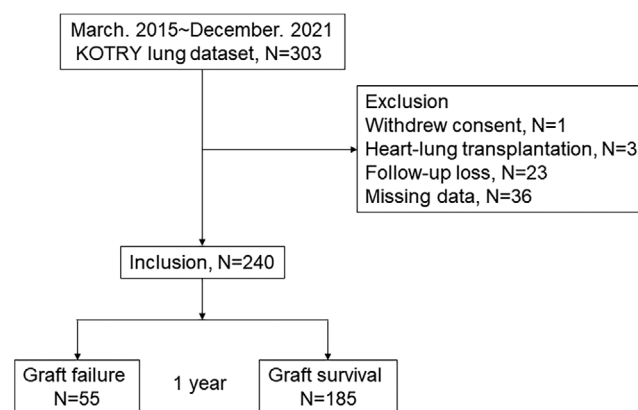


FIGURE 1 | Patient inclusion during the study period, a total of 303 patients were initially enrolled. Patients who withdrew from the registry, underwent heart-lung transplantation, were lost to follow-up, or had incomplete data were excluded from the study. After applying these exclusion criteria, 240 patients remained eligible and were included in the final analysis.

positive blood culture prior to lung transplantation. Pretransplant infections were defined as any clinically diagnosed active infection of any etiology (bacterial, viral, or fungal) requiring antimicrobial therapy prior to transplantation. Colonization—defined as a positive culture without accompanying clinical signs or symptoms and requiring no treatment—was not considered an infection. Pretransplant corticosteroid use was defined as systemic corticosteroid therapy at a dose of ≥ 20 mg/day of prednisone equivalent for at least 2 consecutive weeks prior to transplantation. Status before lung transplantation was based on the urgency classification in the South Korean lung allocation system, which categorizes candidates into Status 0 to 4 according to respiratory support requirements and disease severity [7]. Status 0 indicates highly urgent cases, Status 1 as urgent, and Status 2 or higher as elective. Detailed criteria are provided in Table S1. Chronic kidney disease (CKD) was defined as an estimated glomerular filtration rate (eGFR) of < 60 mL/min/1.73 m², calculated using the CKD-EPI formula, and persisting for ≥ 3 months before transplantation. Cardiac disease was defined as a documented history of ischemic heart disease (including myocardial infarction, prior percutaneous coronary intervention, or coronary artery bypass grafting), heart failure with reduced ejection fraction (left ventricular ejection fraction $< 40\%$), or moderate to severe valvular heart disease as determined by echocardiography. Other clinical and laboratory variables were defined based on values recorded at the time of transplant registration or on the day of transplantation. For laboratory measures such as serum creatinine, the most recent value on the day of transplant was used. When multiple preoperative values were available, only the measurement closest to the time of transplant was included.

2.2 | Machine Learning-Based Model Development

We utilized six ML models to predict 1-year graft failure after lung transplantation, including logistic regression (LR) [8], support vector machine (SVM) [9], random forest (RF) [10], gradient

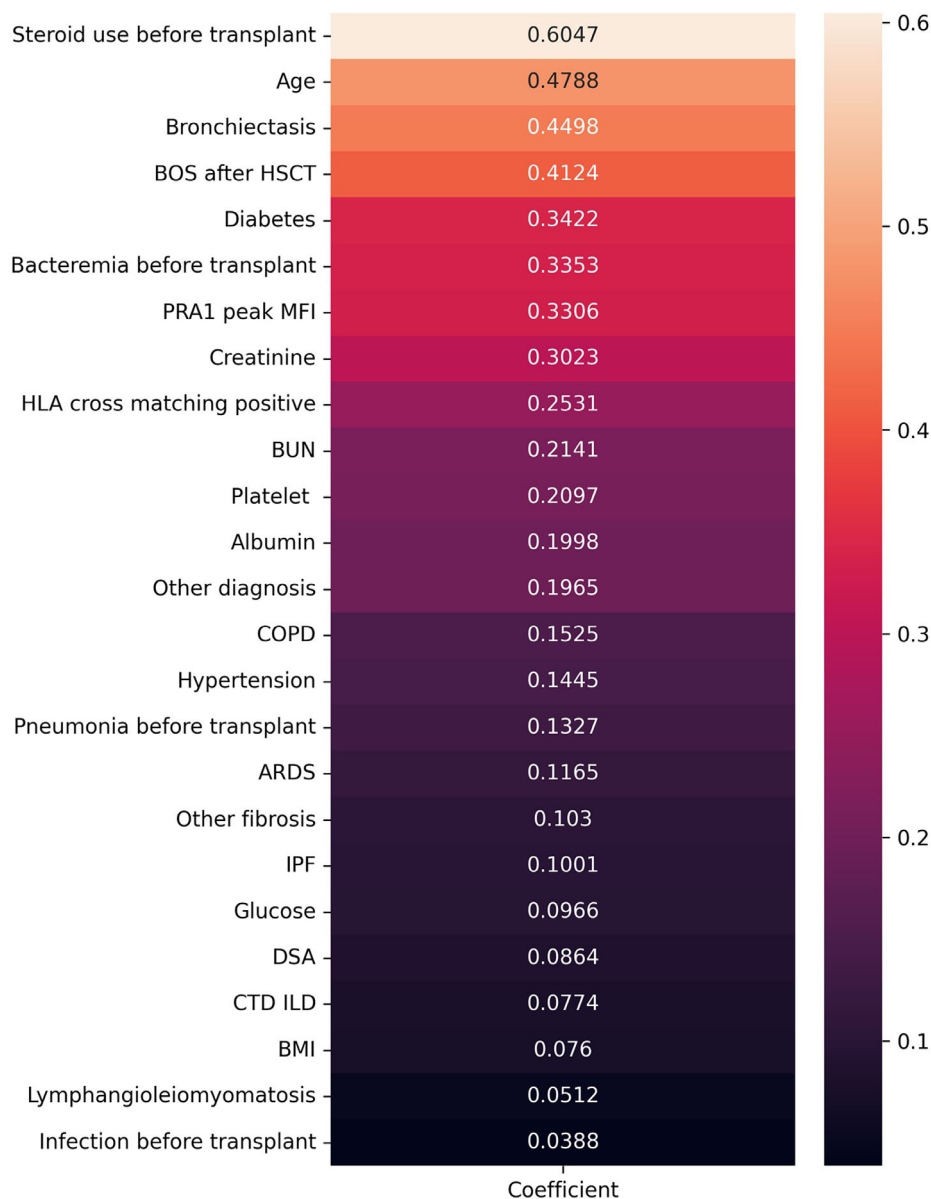


FIGURE 2 | Coefficients of logistic regression model for predicting 1-year graft failure. The analysis focused on identifying the most influential preoperative factors, with only 9 variables having coefficients greater than 0.25 selected from the 25 major preoperative factors.

boosting machine (GBM) [11], Balanced Random Forest (BRF) [12], and Multilayer Perceptron (MLP) [13]. Among the preoperative variables, 25 features that showed significant associations with 1-year graft failure in univariate analysis were selected for model development. These variables are presented in Figure 2. Variables reflecting intraoperative or early postoperative outcomes were excluded from model training, as the goal was to develop a predictive model based solely on pretransplant information. The model coefficients were averaged over five iterations of 5-fold cross-validation to ensure robustness. Based on these averaged coefficients, we selected the top 9 variables with normalized absolute coefficients exceeding 0.25 for model refinement and final development (Figure 3). The nine key factors are as follows: age, bronchiolitis obliterans syndrome (BOS) after hematopoietic stem cell transplantation (HSCT), bacteremia before transplant, bronchiectasis, creatinine, diabetes, human leukocyte antigen (HLA) cross matching positive, panel reactive

antibody (PRA) 1 peak mean fluorescence intensity (MFI), Steroid use before transplant. To verify the appropriateness of this feature selection strategy, we performed additional experiments using alternative approaches such as SHapley Additive exPlanations (SHAP) value-based feature selection (especially from the MLP model), variance thresholding, and clustering-based methods. However, LR-based feature selection consistently yielded the best and most stable performance across all models. Therefore, we applied this same 9-feature set to all ML models for consistent comparison and fair evaluation.

To address class imbalance, we used class weighting during model training. Specifically, the `class_weight = "balanced"` parameter from the scikit-learn package [14] was applied to LR, RF, and SVM models. The BRF model was implemented using the imbalanced-learn package [15], and the GBM model did not support class weighting. Each model was first evaluated using default

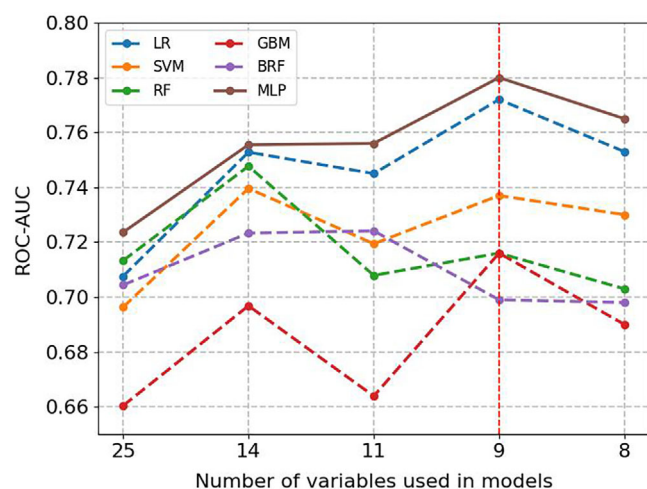


FIGURE 3 | Prediction performance of 1-year graft failure. According to the number of variables used in models. Feature selection was performed by sequentially including variables based on their normalized coefficients from the initial MLP model (as shown in Figure 2), rather than using random subsets. We tested model performance (AUC) as the number of top-ranked features increased from 1 to 25. The highest predictive performance was observed when using 9 features, at which point the lowest included coefficient exceeded 0.25. This configuration was chosen for the final model.

hyperparameters. Subsequently, hyperparameter optimization was performed using grid search. For the SVM model, custom hyperparameters were applied, including an RBF kernel with $C = 0.4$ and $\gamma = \text{"scale"}$. The default settings were retained for the other models, as tuning did not lead to improvement.

The MLP model was built using the Keras library [16] with a three-layer architecture: the first and second Dense layers contained 60 units each with ReLU activation and a dropout rate of 0.7, and the output layer had a single unit with sigmoid activation for binary classification. Due to the limited dataset size, we used 5-fold cross-validation for model evaluation, reporting results as the average performance across folds. To ensure consistent class distribution, the folds were stratified by the outcome (graft failure), preserving its proportion in each fold. We also repeated the analysis using 10-fold cross-validation, which yielded comparable results (Table S2). To interpret the MLP model predictions, which showed the best performance among the six models, we applied the DeepExplainer method from the SHAP Python package [17]. SHAP values were calculated separately within each fold of the 5-fold cross-validation, and the final feature importance results were obtained by averaging SHAP values across folds to ensure robustness.

3 | Statistical analysis

Clinical variables were presented as mean $\pm$ standard deviation or median [interquartile range] for continuous variables and as frequency (percentage) for categorical variables. Categorical variables were compared using the chi-squared test or Fisher's exact test, as appropriate. Survival analysis was conducted using the Kaplan–Meier method, with comparisons made using the log-rank test. The Kaplan–Meier method was conducted using

the lifelines Python package [17]. All other statistical analyses were carried out using R software (version 4.2.3; R Foundation for Statistical Computing, Vienna, Austria), and p values less than 0.05 were considered statistically significant. Only patients with complete data for all selected preoperative variables were included in the analysis. Consequently, there were no missing values in the dataset used for model development, and no imputation was required.

4 | Results

4.1 | Patient Characteristics

Of the 240 patients analyzed, 185 survived for 1 year, 46 died, and 9 underwent retransplantation due to graft failure. As a result, a total of 55 patients were included in the graft failure group (Figure 1). The preoperative characteristics of the two groups are shown in Table 1. Pretransplant steroid use was more frequent in the graft failure group compared to the survival group (63.2% vs. 89.1%, $p < 0.001$). Diabetes mellitus was also more prevalent in the graft failure group (19.5% vs. 38.2%, $p = 0.004$). Pretransplant infections were higher in the graft failure group (21.1% vs. 36.4%, $p = 0.021$), particularly pneumonia (19.5% vs. 32.7%, $p = 0.039$) and bacteremia (1.1% vs. 7.3%, $p = 0.010$). The mean albumin level was lower in the graft failure group (3.4 vs. 3.1 g/dL, $p = 0.004$). HLA cross-matching positivity was more frequent in the graft failure group (2.7% vs. 9.1%, $p = 0.037$). Additionally, the median PRA I peak MFI value was higher in the graft failure group (1000 [549.3–17653] vs. 1949.5 [798.3–3036.5], $p = 0.036$).

4.2 | Operation Related Characteristics and Early Postoperative Outcomes of Lung Transplant Recipients

Bilateral lung transplantation was performed in 96.3% of recipients. There were no significant differences in ischemic time or intraoperative blood loss between the two groups (Table 2). Primary graft dysfunction (PGD) occurred in 34.6% of recipients, with no significant difference between the groups. Postoperative bleeding was more frequent in the graft failure group (14.6% vs. 34.5%, $p = 0.001$). Postoperative infections were also higher in the graft failure group (23.8% vs. 41.8%, $p = 0.009$). Wound complications were more common in the graft failure group (0.5% vs. 5.5%, $p = 0.012$). The median intensive care unit (ICU) stay was longer in the graft failure group (8 [6–15] vs. 14 [8–34] days, $p = 0.001$). Postoperative extracorporeal membrane oxygenation (ECMO) use was more frequent in the graft failure group (49.7% vs. 65.5%, $p = 0.040$), and the median duration of ECMO was also longer (0 [0–2] vs. 2 [0–4] days, $p = 0.002$). Postoperative continuous renal replacement therapy (CRRT) use was significantly higher in the graft failure group (8.1% vs. 41.8%, $p < 0.001$).

Survival to discharge was achieved in 87.5% of recipients, with notable differences in discharge outcomes. A higher proportion of survivors were fully independent at discharge (38.5% vs. 10.7%, $p = 0.001$). Discharge to home was more frequent among survivors (96.2% vs. 75.0%, $p < 0.001$). Tracheostomy at discharge was more commonly maintained in the graft failure group (6.0% vs. 17.9%, $p = 0.028$).

TABLE 1 | Baseline characteristics of lung transplant recipients.

	Total (n = 240)	Graft survival (n = 185)	Graft failure (n = 55)	p value
Age	56.1 ± 9.5	55.5 ± 9.6	58.1 ± 9.2	0.079
Male	158 (65.8)	122 (65.9)	36 (65.5)	0.946
Height	164.9 ± 8.4	165.0 ± 8.8	164.4 ± 7.0	0.570
Weight	58.1 ± 11.9	57.8 ± 12.1	59.1 ± 11.6	0.495
BMI	21.3 ± 3.7	21.1 ± 3.6	21.9 ± 4.1	0.214
Diagnosis				0.482
IPF	131 (54.6)	101 (54.6)	30 (54.5)	
Other fibrosis	9 (3.8)	7 (3.8)	2 (3.6)	
COPD	9 (3.8)	7 (3.8)	2 (3.6)	
CTD-ILD	41 (17.1)	34 (18.4)	7 (12.7)	
Idiopathic pulmonary artery hypertension	1 (0.4)	1 (0.5)	0	
Bronchiectasis	10 (4.2)	10 (5.4)	0	
Lymphangioleiomyomatosis	1 (0.4)	1 (0.5)	0	
BOS after HSCT	16 (6.7)	10 (5.4)	6 (10.9)	
ARDS	16 (6.7)	10 (5.4)	6 (10.9)	
Other	10 (4.2)	7 (3.8)	3 (5.5)	
PFT				
FEV1 (%)	45.3 ± 18.8	45.4 ± 17.4	45.0 ± 23.6	0.916
FVC (%)	44.5 ± 16.1	44.9 ± 14.6	42.9 ± 21.1	0.610
6MWT, Distance (m)	263.8 ± 154.1	271.8 ± 156.7	235.5 ± 145.0	0.354
LV EF (%)	63.8 ± 8.7	64.3 ± 7.8	62.4 ± 10.9	0.185
Mean PAP (mmHg) ^a	25 [21.3–30.8]	25.5 [22–30.8]	24.5 [21–34]	0.381
Steroid use before transplant	166 (69.2)	117 (63.2)	49 (89.1)	< 0.001
Status before transplantation ^b				0.647
Status 0	121 (50.4)	90 (48.6)	31 (56.4)	
Status 1	100 (41.7)	81 (43.8)	19 (34.5)	
Status 2–4	19 (7.9)	14 (7.6)	5 (9.1)	
ICU stay before transplantation (days) ^a	2 [0–16]	1 [0–14]	3 [0–23]	0.502
MV before transplantation	110 (45.8)	80 (43.2)	30 (54.5)	0.140
MV duration (days) ^a	17 [6–32]	15 [5–29]	22.5 [6.8–40]	0.359
ECMO before transplantation	86 (35.8)	65 (35.1)	21 (38.2)	0.679
ECMO duration (days) ^a	0 [0–6.8]	0 [0–6]	0 [0–9]	0.679
Diabetes	57 (23.8)	36 (19.5)	21 (38.2)	0.004
Hypertension	50 (20.8)	36 (19.5)	14 (25.5)	0.336
Chronic kidney disease	2 (0.8)	1 (0.5)	1 (1.8)	0.360
Osteoporosis	33 (13.8)	28 (15.1)	5 (9.1)	0.253
Malignancy history	36 (15.0)	26 (14.1)	10 (18.2)	0.452
Cerebrovascular disease	14 (5.8)	10 (5.4)	4 (7.3)	0.604
Cardiac disease	41 (17.1)	30 (16.2)	11 (20.0)	0.513
Other organ transplant history	18 (7.5)	12 (6.5)	6 (10.9)	0.274
Infection before transplantation	59 (24.6)	39 (21.1)	20 (36.4)	0.021

(Continues)

TABLE 1 | (Continued)

	Total (n = 240)	Graft survival (n = 185)	Graft failure (n = 55)	p value
Pneumonia before transplantation	54 (22.5)	36 (19.5)	18 (32.7)	0.039
Bacteremia before transplantation	6 (2.5)	2 (1.1)	4 (7.3)	0.010
Creatinine	0.6 [0.4–0.7]	0.6 [0.4–0.7]	0.6 [0.4–0.9]	0.898
Albumin	3.4 ± 0.6	3.4 ± 0.6	3.1 ± 0.6	0.004
CMV Ig G positive ^c	229 (99.1)	177 (98.9)	52 (100)	0.444
HLA cross-matching positive	10 (4.2)	5 (2.7)	5 (9.1)	0.037
DSA positive before transplantation	27 (11.3)	18 (9.7)	9 (16.4)	0.172
PRA I peak MFI ^a	1188.5 [632.8–2435.8]	1000 [549.3–17653]	1949.5 [798.3–3036.5]	0.036
PRA II peak MFI ^a	2384 [1053–4090.5]	2384 [1000–3774]	2198.5 [1123–8282.8]	0.920

Note: Data are presented as mean ± SD or n (%), as appropriate.

Abbreviations: ARDS, acute respiratory distress syndrome; BMI, body mass index; BOS after HSCT, bronchiolitis obliterans syndrome after hematopoietic stem cell transplantation; CMV, cytomegalovirus; COPD, chronic obstructive pulmonary disease; CTD-ILD, connective tissue disease-related interstitial lung disease; DSA, donor-specific antibodies; ECMO, extracorporeal membrane oxygenation; FEV1, forced expiratory volume in 1 s; FVC, forced vital capacity; 6MWT, 6 6-minute walk test; HLA, Human Leukocyte Antigen; ICU, Intensive Care Unit; IPF, idiopathic pulmonary fibrosis; LV EF, left ventricular ejection fraction; MFI, mean fluorescence index; MV, Mechanical ventilation; PAP, pulmonary arterial pressure; PFT, pulmonary function test; PRA, panel reactive antibody.

^aData are expressed as median [IQR].

^bThis classification is based on clinical urgency. Patients were categorized into Status 0, 1, 2, 3, or 4 according to medical urgency, with Status 0 defined as highly urgent, Status 1 as urgent, and Status 2 or higher as elective. Detailed criteria are provided in Table S1.

^cN = 229: survival group, n = 179; death group, n = 52.

4.3 | Performance of Six Models for Predicting 1-Year Graft Failure

The performance of the model for 1-year graft failure after lung transplantation is summarized in Table 3. Model performance was assessed using seven metrics: area under the receiver operating characteristic curve (AUC), accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and fl. The six models exhibited performance ranging from 0.699 to 0.780. Among the six models we implemented, the MLP demonstrated the highest performance. Despite the limited size of the training dataset, the MLP exhibited acceptable predictive capability. Figure 4 shows the feature importance analysis using SHAP. “Steroid use before transplant” had the greatest influence on the MLP’s predictions of 1-year graft failure, followed by age, diabetes, BOS after HSCT, and bronchiectasis (Figure 4A). The model classified individuals as deceased if they had a higher age, a history of steroid use before transplantation, diabetes, BOS after HSCT, and the absence of bronchiectasis (Figure 4B).

4.4 | Real Clinical Outcomes of the Group Predicted to Experience Graft Failure by the MLP Model

To further evaluate the clinical utility of the model, we conducted a survival analysis based on the MLP model’s predictions. Patients were stratified into “Low-risk” and “High-risk” groups, where the Low-risk group included those predicted to achieve graft survival, and the High-risk group included those predicted to experience graft failure. The survival outcomes between the two

groups differed significantly ($\chi^2 = 49.85$, $p < 0.001$, Figure 5). In the predicted survival group, only 11.5% of patients experienced graft failure, whereas 44.6% of those in the predicted graft failure group did.

Additionally, we compared early postoperative outcomes between patients classified as Low-risk and those classified as High-risk by the MLP model. There was no significant difference in the incidence of PGD between the two groups (37.3% vs. 29.1%, $p = 0.212$) or in the use of postoperative ECMO (53.4% vs. 53.2%, $p = 0.971$). However, postoperative CRRT was significantly more frequent in the predicted graft failure group (24.1% vs. 11.8%, $p = 0.015$). Airway complications were also significantly higher in this group (6.3% vs. 0.6%, $p = 0.008$). Postoperative infections showed a trend toward being more common in the predicted graft failure group (35.4% vs. 24.2%, $p = 0.069$), as did postoperative bleeding (25.3% vs. 16.1%, $p = 0.090$). There were no significant differences in other complications between the two groups. Moreover, the median postoperative ICU stay was significantly longer in the predicted graft failure group (12 [6–22] vs. 8 [6–14] days, $p = 0.032$), as was the duration of postoperative mechanical ventilation (8 [5–24] vs. 5 [4–9] days, $p = 0.004$).

5 | Discussion

In this study, we developed a ML-based model to predict 1-year graft failure in South Korean lung transplant recipients using only pretransplant factors. Among 240 patients, 55 (22.9%) experienced graft failure within 1 year. Our MLP model demonstrated strong predictive performance (AUC = 0.780). This suggests that

TABLE 2 | Operation-related characteristics and Early post-operative outcomes of lung transplant recipients

	Total (n = 240)	Graft survival (n = 185)	Graft failure (n = 55)	p value
Type of operation				0.449
Bilateral lung transplantation	231 (96.3)	179 (96.8)	52 (94.5)	
Single lung transplantation	9 (3.8)	6 (3.2)	3 (5.5)	
Total ischemic time (minutes)	601.3 ± 182.3	593.2 ± 182.4	629.4 ± 180.6	0.208
Warm ischemic time (minutes)	135.2 ± 78.7	127.2 ± 48.2	162.9 ± 137.2	0.071
Estimated volumes of blood loss (mL) ^a	3000 [1400–7500]	2700 [1250–7000]	3200 [2000–9999]	0.377
PGD	83 (34.6)	66 (35.7)	17 (30.9)	0.514
Postoperative bleeding	46 (19.2)	27 (14.6)	19 (34.5)	0.001
Airway complications	6 (2.5)	3 (1.6)	3 (5.5)	0.110
Vascular complications	2 (0.8)	2 (1.1)	0	0.439
Pulmonary thromboembolism	2 (0.8)	2 (1.1)	0	0.439
Infection	67 (27.9)	44 (23.8)	23 (41.8)	0.009
Pneumonia	51 (21.3)	35 (18.9)	16 (29.1)	0.105
Wound problems	4 (1.7)	1 (0.5)	3 (5.5)	0.012
Postoperative ICU stay (days)*	9 [6–16]	8 [6–15]	14 [8–34]	0.001
Postoperative MV days ^a	6 [4–14.8]	6 [4–11]	8 [5–25]	<0.001
Postoperative ECMO	128 (53.3)	92 (49.7)	36 (65.5)	0.040
Postoperative ECMO days ^a	1 [0–3]	0 [0–2]	2 [0–4]	0.002
Postoperative CRRT	38 (15.8)	15 (8.1)	23 (41.8)	<0.001
Hospital survival	210 (87.5)	182 (98.4)	28 (50.9)	<0.001
BMI at discharge	19.6 ± 3.1	19.7 ± 3.1	19.4 ± 3.5	0.673
Functional status at discharge (total N = 210)				0.001
Fully independent	73 (34.8)	70 (38.5)	3 (10.7)	
Partially dependent	125 (59.5)	105 (57.7)	20 (71.4)	
Fully dependent	12 (5.7)	7 (3.8)	5 (17.9)	
Home discharge (total N = 210)	196 (93.3)	175 (96.2)	21 (75.0)	<0.001
Tracheostomy at discharge (total N = 210)	16 (7.6)	11 (6.0)	5 (17.9)	0.028
Home oxygen therapy (total N = 210)	6 (2.9)	6 (3.3)	0	0.330

Note: Data are presented as mean ± SD or n (%), as appropriate.

Abbreviations: BMI, body mass index; CRRT, continuous renal replacement therapy; ECMO, extracorporeal membrane oxygenation; ICU, intensive care unit; MV, mechanical ventilator; PGD, primary graft dysfunction.

^aData are expressed as median [IQR].

TABLE 3 | Performance of six models for predicting 1-year graft failure

Models	AUC	Accuracy	sensitivity	specificity	PPV	NPV	f1
LR	0.774	0.704	0.693	0.710	0.426	0.886	0.520
RF	0.716	0.779	0.291	0.924	0.549	0.814	0.375
SVM	0.764	0.721	0.713	0.725	0.442	0.897	0.535
GBM	0.716	0.771	0.345	0.897	0.510	0.823	0.404
BRF	0.699	0.733	0.527	0.795	0.436	0.850	0.474
MLP	0.780	0.733	0.673	0.751	0.449	0.886	0.537

Abbreviations: AUC, area under the curve; BRF, balanced random forest; GBM, gradient boosting machine; LR, logistic regression; MLP, multilayer perceptron; NPV, negative predictive value; PPV, positive predictive value; RF, random forest; SVM, support vector machine.

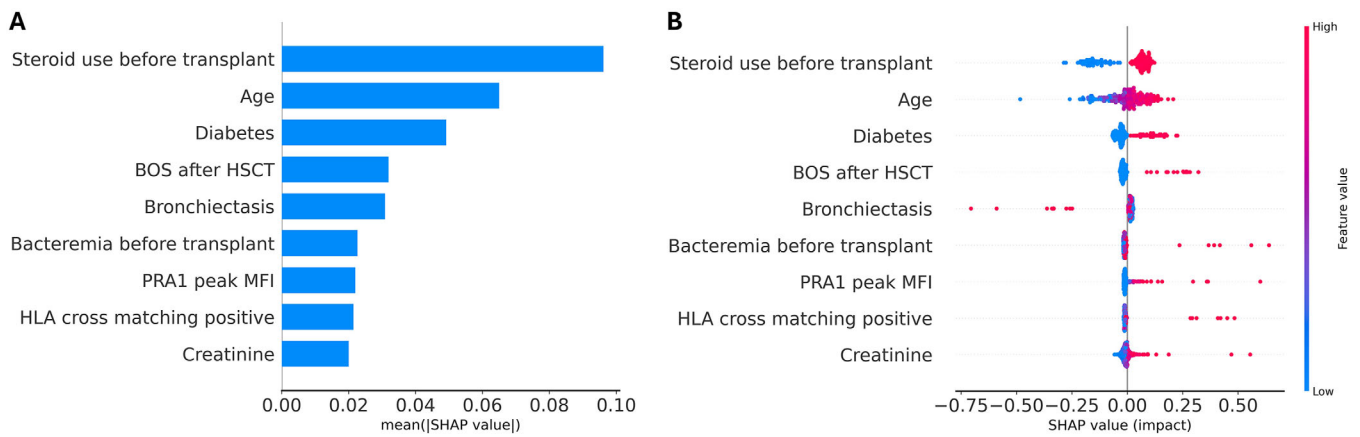


FIGURE 4 | SHAP feature importance and summary plot in predicting 1-year graft failure. SHAP feature importance (A) shows the importance of features in predicting 1-year graft failure using the MLP model. The SHAP summary plot (B) illustrates the impact of each feature on the prediction. For features like Age, where higher values correspond to higher SHAP values, it can be interpreted that higher age has a greater influence on the prediction of graft failure. In contrast, the presence of bronchiectasis (red) is associated with a lower predicted risk of graft failure.

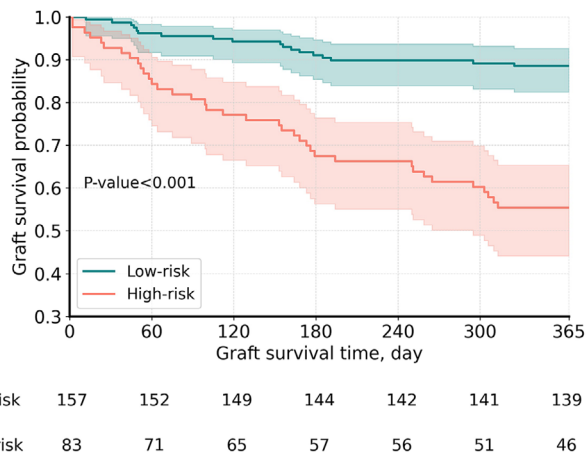


FIGURE 5 | Kaplan-Meier curves for actual graft failure in two risk groups classified by the Model. In the low-risk group, only 11.5% of patients experienced graft failure, compared to 44.6% in the high-risk group.

the model effectively differentiates between high-risk and low-risk patients, potentially aiding in clinical decision-making and transplant candidate selection.

Previous models have incorporated recipient, donor, and perioperative factors, including postoperative complications such as bleeding [1, 18]. In contrast, our model focuses solely on pretransplant recipient characteristics, making it more practical for use at the time of transplant evaluation. We identified nine key predictors of 1-year graft failure, with pretransplant steroid use emerging as the most influential, followed by age, diabetes, bronchiectasis, and bacteremia. Notably, infection-related factors had a greater impact on outcomes than immunologic variables such as HLA mismatch or PRA levels. This aligns with the high infection burden observed in South Korean lung transplant recipients, where infection accounted for 58.2% of 1-year graft failures—far exceeding refractory rejection (7.3%) (Table S3) [19]. These findings emphasize the need for robust perioperative infection prevention and management strategies. Meanwhile, stratified analysis of model performance by cause of death

revealed relatively high predictive accuracy for nonpneumonia infections and graft failure of unknown cause but relatively lower accuracy for pneumonia-related deaths. (Table S3). This suggests that pneumonia may be harder to predict due to its clinical heterogeneity, underscoring a key area for model refinement.

The strong association between pretransplant steroid use and graft failure is particularly noteworthy. Nearly 70% of recipients in our cohort were on steroids prior to transplant. While steroids are often necessary for managing chronic lung disease, prolonged use can impair immune function and tissue healing, increasing the risk of infection and surgical complications [20]. These findings suggest that minimizing unnecessary steroid use in end-stage lung disease may improve posttransplant outcomes.

Interestingly, the recipient's underlying disease was not a major predictor of 1-year graft failure—unlike prior studies using UNOS data [21]. Of the ten diagnostic categories, only BOS following HSCT was associated with increased risk, while bronchiectasis was associated with better outcomes. This implies that treatment-related factors, rather than diagnosis itself, may play a greater role in posttransplant prognosis. Differences in patient characteristics between South Korean and Western populations, including higher acuity at transplant and frequent use of ECMO as a bridge due to donor shortages, may also contribute to this discrepancy. Further multicenter studies with larger cohorts are needed to validate these findings.

Beyond survival, functional recovery after lung transplantation remains a major concern. In our study, while 93.3% of hospital survivors were discharged home, only 34.8% were fully independent at the time of discharge. Furthermore, among those discharged home, 10.7% required retransplantation or died within 1 year due to complications. These findings highlight the substantial burden of lung transplantation and the importance of careful candidate selection, long-term posttransplant management, and rehabilitation strategies. A key challenge in lung transplant decision-making is the ethical dilemma surrounding candidate selection and delisting decisions. In South Korea, idiopathic pulmonary fibrosis (IPF) is a leading indication for lung transplantation,

and due to limited donor availability, many patients require ECMO as a bridge to transplant. In such critically ill patients, clinicians often face substantial pressure when deciding whether to proceed with transplantation or to delist candidates—decisions that frequently determine survival. These high-stakes choices are largely based on clinical judgment, which may introduce variability and potential bias in patient selection.

ML offers a promising solution to enhance objectivity and consistency in transplant decision-making. Unlike traditional risk assessment tools, ML models can integrate multiple pretransplant variables to provide personalized risk predictions. Compared to recent ML models developed using UNOS data, which demonstrated limited predictive performance despite incorporating larger datasets (AUC = 0.63–0.64 for XGBoost and EHRFormer), our model achieved relatively strong performance (AUC = 0.780) despite utilizing a smaller dataset with fewer variables [22]. This suggests that a well-selected set of pretransplant factors may offer valuable prognostic insights, even in resource-limited settings. Given the comparable performance of LR and MLP models, the trade-off between predictive performance and interpretability should be considered for clinical application. LR allows variable importance to be expressed as odds ratios familiar to clinicians, which may facilitate acceptance and implementation. However, even modest gains in predictive accuracy observed with the MLP may translate into meaningful clinical benefits, particularly in larger populations. Additionally, deep learning models such as MLPs tend to perform better with larger datasets and may offer greater predictive gains as future data availability increases. Therefore, while LR remains a practical and interpretable option, retaining more complex ML models like MLP could provide valuable incremental improvements in predictive performance.

Despite its promising performance, our study has several limitations. First, the model was developed using a relatively small cohort of South Korean lung transplant recipients. This may limit its generalizability to non-South Korean populations, such as those represented in UNOS or other Western registries. However, our primary aim was to develop a model optimized for the South Korean lung transplant setting. To confirm its clinical utility and robustness within this context, prospective external validation using independent South Korean cohorts is planned. Second, although our model outperformed previous ML models based on UNOS data, further optimization is needed. Exploring alternative ML approaches such as ensemble learning, deep learning, or hybrid models integrating structured and unstructured clinical data (e.g., imaging or pathology reports) could enhance predictive performance. Third, a small proportion of deaths (1.8%) and graft failures (9.1%) were categorized as having an unknown cause. While these cases were included in the analysis under the definition of all-cause graft failure, the absence of detailed cause-of-death data limits the interpretability of specific outcome patterns. In South Korea, where autopsies are not routinely performed, this remains a common limitation of real-world transplant data. Fourth, our model consistently demonstrated higher NPV than PPV, a pattern also reported in other lung allocation and survival models. One explanation may be the use of class weighting, which increases sensitivity but may lead to a higher rate of false positives. In contrast, negative predictions were likely made with greater certainty, resulting in elevated NPV. This effect was more pronounced in the setting of class imbalance, where the number

of survivors significantly exceeded the number of deaths. While high NPV aligns with the clinical goal of identifying candidates with favorable prognosis, the relatively lower PPV underscores a limitation in detecting patients at high risk for graft failure. Future work should explore cost-sensitive learning or calibration strategies to improve sensitivity for adverse outcomes. Future research should prioritize prospective validation in real-world clinical settings. Incorporating real-time patient data and refining predictive algorithms over time may enhance candidate selection and posttransplant management. Additionally, developing a clinically interpretable decision-support system that integrates with electronic health records could facilitate timely, data-driven decisions by transplant teams. Ultimately, this study underscores the potential of ML to improve lung transplant decision-making, optimize resource allocation, and personalize care—contributing to improved long-term outcomes for transplant recipients.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Table S1: The Registration Criteria for a Lung Transplant in Korea.

Table S2: Comparison of model performance between 5-fold and 10-fold cross-validation.

Table S3: Model accuracy stratified by cause of death among patients who experienced graft failure.